

MERN Stack Syllabus

Duration: 12 Weeks (3 Months)

Tech Stack: MongoDB, Express.js, React.js, Node.js

Extras Included: Redux Toolkit, Git & GitHub, Free Hosting, REST APIs, Authentication, Mini Projects, Major Project

Month 1 — Frontend Development & Programming Foundations

Week 1 — Web Basics & HTML/CSS

Topics

- Internet & How Websites Work
- VS Code Setup
- HTML5 Basics
- Semantic Tags
- Forms & Inputs
- CSS Basics
- Flexbox
- CSS Grid
- Responsive Design
- Media Queries

Mini Project

- Personal Portfolio Website

Week 2 — Advanced CSS, Tailwind CSS & Git/GitHub

Topics

CSS & Tailwind

- CSS Animations
- Transitions
- Positioning
- Tailwind CSS Introduction
- Responsive Tailwind
- Reusable UI Components
- Modern Layout Design

Git & GitHub

- Introduction to Git
- Installing Git
- Git Commands
- Repository Creation
- GitHub Basics
- Commit & Push
- Branching
- Clone & Pull

Mini Project

- Responsive Landing Page + Upload to GitHub

Week 3 — JavaScript Fundamentals

Topics

- Variables & Data Types
- Operators
- Conditions
- Loops
- Functions
- Arrays
- Objects
- DOM Manipulation
- Events
- ES6 Features
 - Arrow Functions
 - Destructuring
 - Spread Operator

Mini Project

- Weather App

Week 4 — Advanced JavaScript

Topics

- Array Methods
- Closures
- Callback Functions
- Promises
- Async/Await
- Fetch API

- Error Handling
- Local Storage
- Session Storage

Mini Project

- Task Manager App

Month 2 — React.js Development

Week 5 — React Fundamentals

Topics

- Introduction to React
- Vite Setup
- JSX
- Components
- Props
- State
- Event Handling
- Conditional Rendering
- Lists & Keys
- Forms in React

Mini Project

- Notes App

Week 6 — React Advanced

Topics

- useEffect Hook
- React Router DOM
- Dynamic Routing
- API Integration
- Context API
- Custom Hooks
- React Folder Structure
- Reusable Components

Mini Project

- Movie Search App

Week 7 — Redux Toolkit

Topics

- State Management
- Redux Fundamentals
- Redux Toolkit
- Store
- Slice
- Reducers
- Actions
- useSelector
- useDispatch

Mini Project

- Shopping Cart System

Week 8 — Frontend Hosting & Deployment

Topics

- Deploying React Apps
- Environment Variables
- Build Optimization
- Free Hosting Platforms
 - [Vercel](#)
 - [Netlify](#)
 - [Render](#)
- Domain Basics

Mini Project

- Deploy Portfolio Website Online

Month 3 — Backend & Full Stack MERN

Week 9 — Node.js & Express.js

Topics

- Node.js Introduction
- NPM
- Modules
- Express.js Basics
- Routes
- Middleware
- REST API Development
- Request & Response
- CRUD Operations

Mini Project

- Student Management API

Week 10 — MongoDB & Authentication

Topics

- MongoDB Basics
- Collections & Documents
- Mongoose
- Schema & Models
- Database Relationships
- JWT Authentication
- Bcrypt Password Hashing
- Login/Register APIs
- Protected Routes

Mini Project

- Authentication System

Week 11 — Full Stack Integration

Topics

- Connecting React with Backend
- Axios
- Authentication Flow
- Role-Based Access
- File Upload Basics
- Form Validation
- Toast Notifications

Mini Project

- Blog Management System

Week 12 — Major Project Development & Deployment

Major Project

- **Gadget Rental Platform**

Project Modules

- User Authentication
- Gadget Listings
- Search & Filtering
- Rental Booking System
- Cart & Checkout
- Rental Duration Selection
- Order Management
- Admin Dashboard
- User Dashboard
- Product Availability Tracking
- Image Upload
- Responsive UI